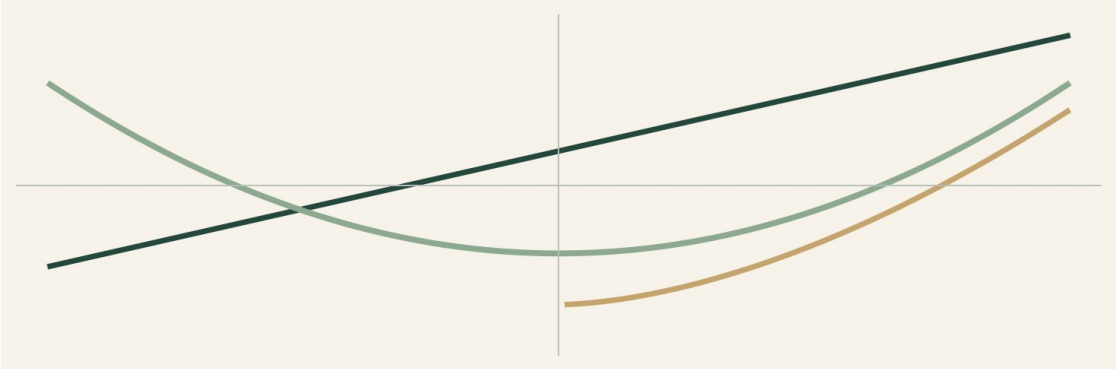
MATHEMATICS • PROJECT REPORT

APPLICATIONS OF STRAIGHT LINES, PARABOLAS & POWER CURVES



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ACADEMIC YEAR 2026–2027

CERTIFICATE AND DECLARATION

Certificate

This is to certify that the project report entitled “Applications of Straight Lines, Parabolas and Power Curves in Real-Life Fields” has been prepared by Harika M (USN: 1EP25CS078), of the Department of Computer Science and Engineering, East Point College of Engineering and Technology, as part of the prescribed academic work.

The work presented in this report is intended to demonstrate the use of mathematical functions in practical situations. The report has been prepared for academic purposes.

Faculty Guide: Ms. Bhanupriya

Faculty signature: _____

Head of Department: _____

Date: _____ Place: _____

Declaration

I hereby declare that this project report has been prepared by me for academic purposes. The explanations and numerical examples are presented to demonstrate mathematical ideas. Sources consulted during preparation are listed in the bibliography, and any illustrative models are identified as examples rather than universal scientific rules.

Student signature: _____

Date: _____

ACKNOWLEDGEMENT

I express my sincere gratitude to my respected Mathematics professor, **Ms. Bhanupriya** , Assistant Professor, Department of Mathematics, for the valuable guidance, encouragement and academic support offered during the preparation of this project. The opportunity to study mathematical functions through practical examples has helped me connect classroom concepts with situations beyond the textbook.

I am grateful to East Point College of Engineering and Technology and to the Department of Computer Science and Engineering (CSE) for providing an environment that encourages curiosity, independent learning and interdisciplinary thinking. The knowledge gained through lectures, class discussions and reference materials has supported the development of this report.

This project gave me an opportunity to explore how the straight line, parabola and power curve can be applied in healthcare, sports, business, engineering, transportation, environmental science and computer science. Preparing the graphs and worked examples strengthened my understanding of mathematical modelling, the interpretation of variables and the importance of checking assumptions.

I also thank the authors and educators whose textbooks and open learning resources helped me understand the theory behind the equations. Their explanations were useful in selecting simple examples that demonstrate how mathematical relationships may be interpreted in everyday contexts.

Finally, I extend my heartfelt thanks to my family, friends and classmates for their encouragement and support throughout the completion of this assignment. Their motivation helped me complete the report with care and enthusiasm.

With sincere gratitude,
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ABSTRACT

Mathematics is an essential tool for understanding the patterns, changes and relationships that appear in daily life. Many situations that seem complex can be represented through relatively simple equations. This project explores three important mathematical functions: the straight line, $y = ax + b$; the parabola, $y = ax^2 + bx + c$; and the power curve, $y = ax^b$. Each function has a distinct shape and describes a different kind of relationship between variables.

A straight-line function represents a constant rate of change and is useful in situations such as estimating travel distance at constant speed, calculating costs with a fixed charge and a per-unit rate, and describing linear trends. A quadratic function produces a parabolic graph and can represent idealized projectile motion, simplified optimization problems and curved trends with a maximum or minimum. A power function describes relationships involving an exponent and is often useful for studying scaling patterns in science, engineering and computing.

The report presents applications from seven different fields: medicine and healthcare, sports and athletics, economics and business, engineering and architecture, transportation and road safety, agriculture and environmental science, and computer science and data analysis. Each field is discussed through short explanations, mathematical expressions, worked examples and illustrative graphs to demonstrate how these functions can be used to describe practical situations.

The study also compares the strengths and limitations of the three models. It emphasizes that mathematical equations are simplified representations of reality, and that assumptions, units, measurement quality and the range of available data must be considered before using any model to make predictions. The graphs provided in the report are illustrative examples rather than experimental observations.

Overall, the project shows that straight lines, parabolas and power curves form a useful foundation for interpreting real-world information. Understanding when and why to use each function improves mathematical reasoning and builds connections between mathematics and other academic disciplines. The report concludes with suggestions for further exploration using real datasets, spreadsheets and interactive graphing tools.

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APPLICATIONS IN DIFFERENT FIELDS

Understanding the Three Functions

A mathematical function describes how one quantity depends on another. In this project, x is the input variable and y is the output variable. The constants determine the shape and position of the graph.

1. Straight line: $y = ax + b$

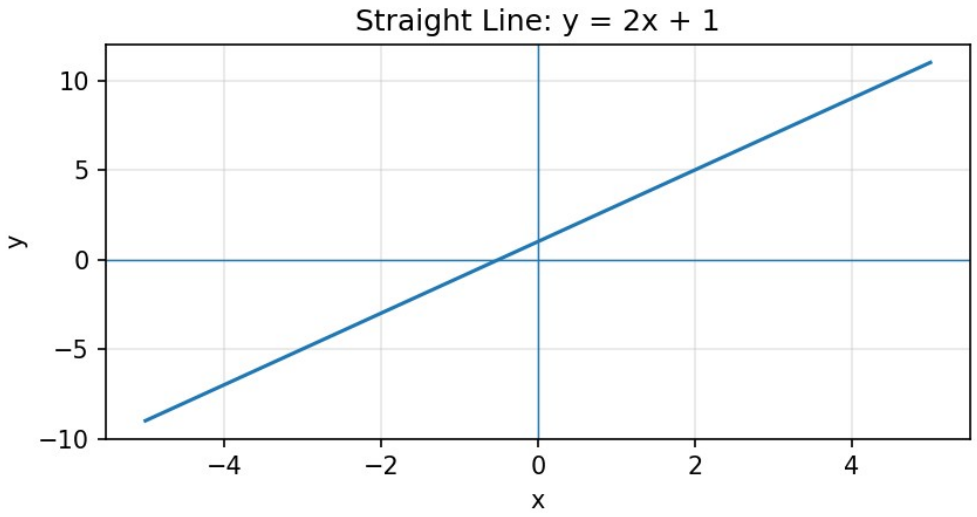
Here, a is the slope or rate of change, and b is the y-intercept. If x increases by one unit, y changes by a units. This model is suitable when the rate of change is approximately constant over the range being studied.

2. Parabola: $y = ax^2 + bx + c$

This is a quadratic function. Its graph is a parabola. If a is positive, the parabola opens upward and has a minimum; if a is negative, it opens downward and has a maximum. The x-coordinate of the vertex is $x = -b/(2a)$, provided a is not zero.

3. Power curve: $y = ax^b$

In a power function, a is a scale factor and b is the exponent. The graph depends on the value of b and the domain of x. Power functions are often used to represent scaling patterns in biology, engineering and computing.



APPLICATION 1: MEDICINE AND HEALTHCARE

Mathematics supports healthcare research by helping researchers calibrate instruments, summarize measurements and investigate patterns in biological data. A mathematical model is an aid to analysis; it does not replace clinical judgement.

A. Straight-line model: instrument calibration

A sensor may produce a reading that is approximately linearly related to a measured quantity. Suppose a simplified calibration relationship is $y = 2x + 5$, where x is the sensor reading and y is the converted value in the instrument's chosen units. If $x = 10$, then $y = 2(10) + 5 = 25$. This demonstrates how a linear equation converts one measurement into another.

B. Parabolic model: approximating a response

Within a limited range, a quadratic function may approximate a response that first rises and then falls as a parameter changes. For example, $y = -x^2 + 6x + 2$ has its vertex at $x = -6/[2(-1)] = 3$. Substitution gives $y = -9 + 18 + 2 = 11$, the model's maximum value. This is a mathematical illustration only, not a treatment recommendation.

C. Power curve: biological scaling

Researchers use relationships of the form $y = ax^b$ to study how biological measurements scale with body size. The constants are estimated from observations, and the exponent depends on the specific measurement and population.

Practical value: these functions can help with measurement, data analysis and research. Any real clinical application requires reliable data, validation and appropriate medical oversight.

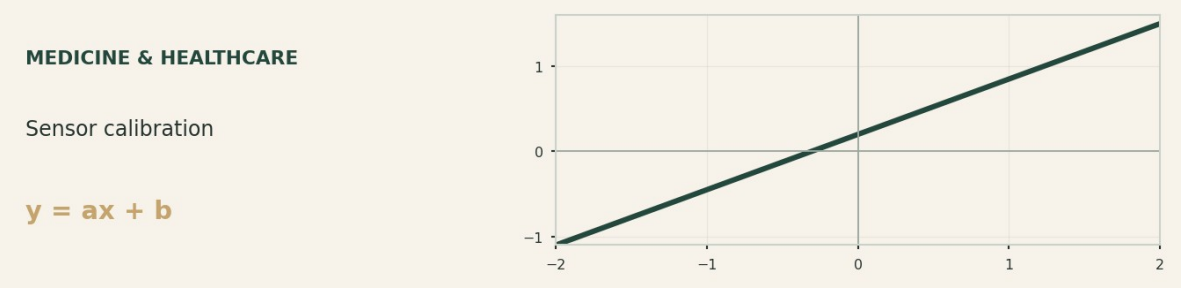


Figure: Illustrative model and graph — not experimental data.

APPLICATION 2: SPORTS AND ATHLETICS

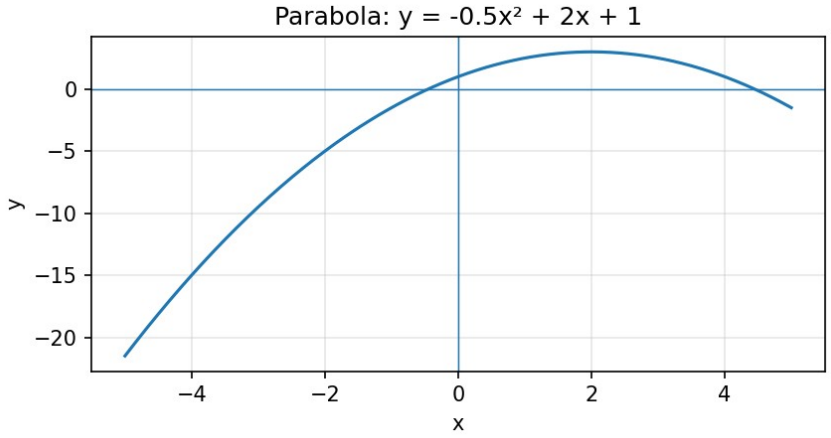
Sports science uses mathematics to examine movement, compare performance and study how equipment travels. The models below simplify real situations to make the mathematical relationship clear.

A. Straight line: constant-speed running

For constant speed, distance is d = vt. If a runner moves at 5 metres per second for 10 seconds, then d = 5 × 10 = 50 metres. Distance is directly proportional to time when speed remains constant, so the graph is a straight line through the origin.

B. Parabola: trajectory of a ball

Ignoring air resistance and assuming a fixed gravitational field, the vertical position of a projectile can be expressed as a quadratic function of horizontal position. Consider the illustrative equation $y = -x^2 + 4x + 1$. At $x = 2$, $y = -4 + 8 + 1 = 5$. The graph opens downward, illustrating a rise followed by a fall.



C. Power curve: scaling performance

Power functions can be fitted to sports data to investigate how certain performance measures relate to body dimensions or other variables. The exponent must be estimated from suitable data; there is no single power law that describes all athletes or sports.

Practical value: models can support motion analysis and performance research, but actual results are affected by technique, fatigue, air resistance and other factors.

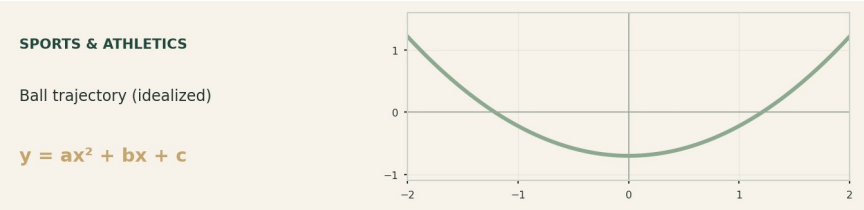


Figure: Illustrative model and graph — not experimental data.

APPLICATION 3: ECONOMICS AND BUSINESS

Businesses use equations to estimate expenses, examine revenue and investigate the output level that may maximize profit.

A. Straight line: total cost

Suppose a small business has a fixed cost of ₹500 and a variable cost of ₹20 per product. Its total cost can be modelled as $C = 20x + 500$. For 30 products, $C = 20(30) + 500 = ₹1{,}100$. The slope 20 represents the additional cost per product, while 500 is the fixed cost.

B. Parabola: simplified profit model

Suppose profit is modelled by $P = -2x^2 + 40x - 100$, where x is the number of units in a suitable range. The vertex occurs at $x = -40/[2(-2)] = 10$. Substituting gives $P = -200 + 400 - 100 = 100$. The model's maximum profit is 100 monetary units at $x = 10$. This conclusion depends on the assumed equation and is not a universal business rule.

C. Power curve: production scaling

A model $C = ax^b$ may describe how a cost changes with production scale when observations show a power-law pattern. If b is below 1, the model grows more slowly than x; if b is above 1, it grows more rapidly. The model must be checked against real business data.

Practical value: mathematical models help compare costs, estimate outcomes and support planning, provided their assumptions are reasonable.

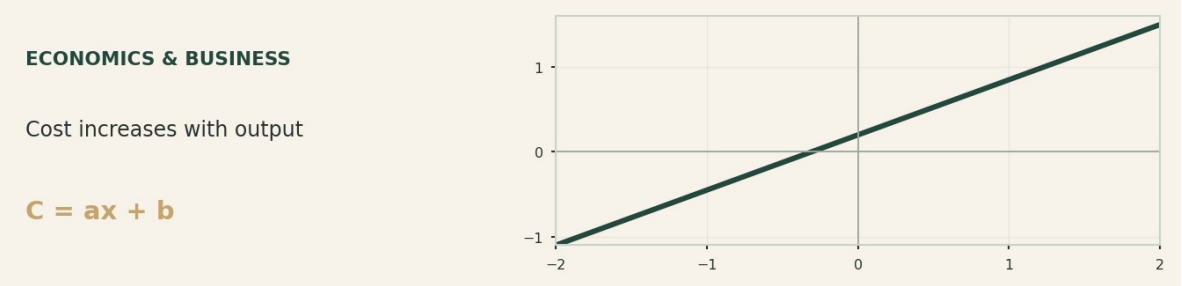


Figure: Illustrative model and graph — not experimental data.

APPLICATION 4: ENGINEERING AND ARCHITECTURE

Engineering uses mathematical functions to describe shapes, estimate dimensions and model how quantities change in a designed system.

A. Straight line: slope and alignment

A straight line can describe a ramp, a road gradient or a linear component in a drawing. For $y = 0.1x + 2$, when $x = 20$, $y = 0.1(20) + 2 = 4$. The slope 0.1 means the height increases by 0.1 unit for every one-unit horizontal increase, if both axes use compatible units.

B. Parabola: reflectors

A parabola is important in the design of parabolic reflectors, such as some satellite dishes and reflecting telescopes. Rays parallel to the axis are reflected toward the focus under ideal geometric conditions. The simple equation $y = x^2$ describes a parabola with vertex at the origin. At $x = 3$, $y = 9$.

C. Power curve: scaling laws

Engineers may use $y = ax^b$ to approximate relationships between size and other quantities, such as mass, energy use or capacity. The exponent depends on the system's geometry and physical properties. Engineers must use appropriate material properties, safety factors and validated design methods before applying such models to real structures.

Practical value: functions help describe designs and relationships, but an equation alone does not certify that a structure is safe.

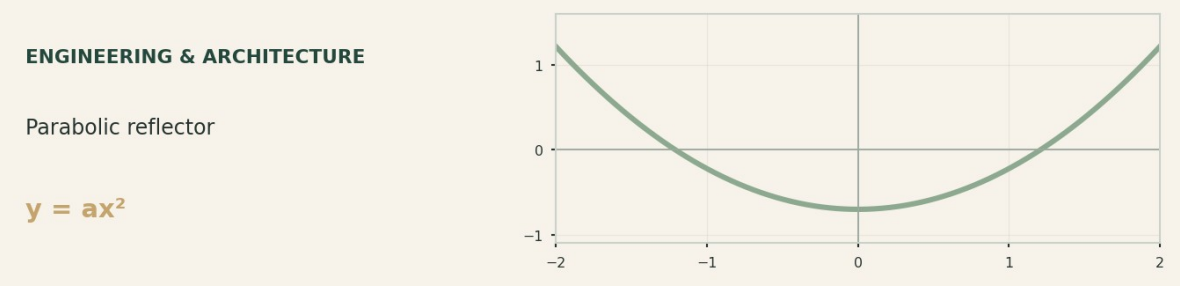


Figure: Illustrative model and graph — not experimental data.

APPLICATION 5: TRANSPORTATION AND ROAD SAFETY

Transportation mathematics helps estimate travel distance, study vehicle motion and understand how speed affects stopping requirements.

A. Straight line: distance travelled

For a vehicle moving at a constant speed of 60 km/h, distance can be written as $d = 60t$, where t is time in hours. In 2.5 hours, $d = 60 \times 2.5 = 150$ km. This is a linear model because the speed is constant.

B. Parabolic relationship: braking distance

Under a simplified model with constant deceleration, braking distance is $d = v^2/(2a)$, where v is the initial speed and a is the positive magnitude of deceleration. If $v = 10$ m/s and $a = 5$ m/s^2, then $d = 100/10 = 10$ m. Because distance depends on the square of speed, doubling speed would quadruple the idealized braking distance if deceleration remained the same.

C. Power curve: transport data

A power model $y = ax^b$ may be fitted to some transport datasets to investigate scaling patterns, such as how a measure of network activity varies with the size of a location. The form is appropriate only when supported by observations.

Practical value: mathematical models help with planning and analysis. Real stopping distances also depend on reaction time, road surface, tyres, weather, vehicle condition and other factors.

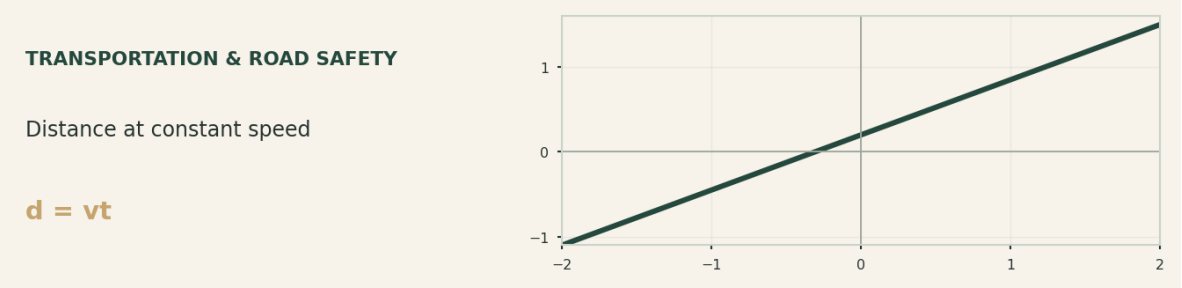


Figure: Illustrative model and graph — not experimental data.

APPLICATION 6: AGRICULTURE AND ENVIRONMENTAL SCIENCE

Agriculture and environmental research use mathematics to estimate resource requirements, analyze plant measurements and investigate trends in crop production.

A. Straight line: water use

Suppose a simplified irrigation model is $W = 3x + 20$, where x is the number of plants and W is water use in chosen units for a given period. For 10 plants, $W = 3(10) + 20 = 50$ units. The equation assumes a fixed baseline and a constant additional amount per plant.

B. Parabola: illustrative crop-yield model

Within a limited experimental range, yield may sometimes be approximated by a quadratic function. Consider $Y = -2x^2 + 20x + 10$, where x represents fertilizer amount in an illustrative scale. The vertex occurs at $x = -20/[2(-2)] = 5$. Substitution gives $Y = -50 + 100 + 10 = 60$. This means the model reaches its maximum at $x = 5$; it is not a real fertilizer recommendation.

C. Power curve: estimating biomass

Ecologists may estimate plant biomass using a model such as $M = aD^b$, where D is a measured dimension such as trunk diameter and M is estimated biomass. The constants are fitted from measurements and can differ by species and growing conditions.

Practical value: these functions can support resource planning and ecological studies. Real outcomes depend on weather, soil, crop variety and management practices.

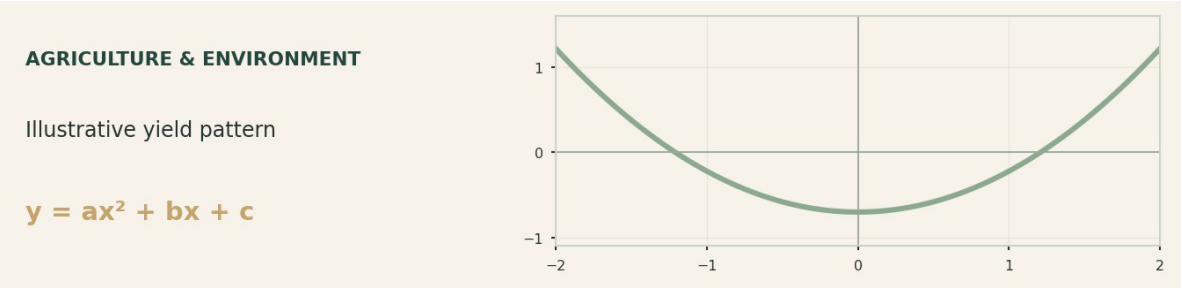


Figure: Illustrative model and graph — not experimental data.

APPLICATION 7: COMPUTER SCIENCE AND DATA ANALYSIS

Computer science uses mathematical functions to fit trends, analyse datasets and estimate the resources needed by algorithms.

A. Straight line: trend modelling

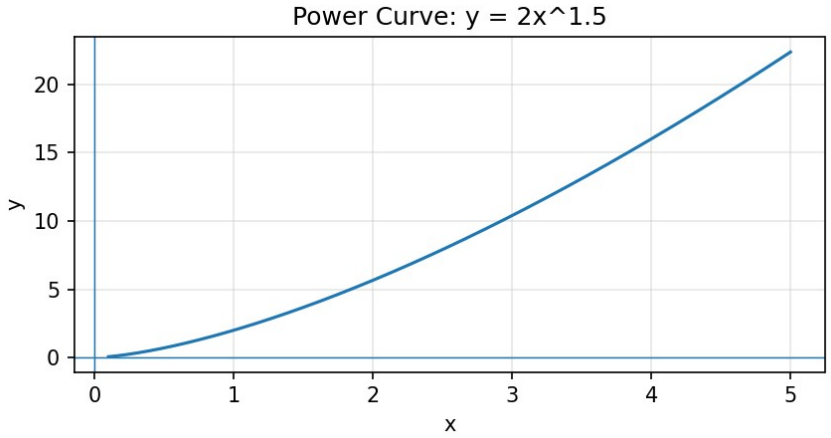
A linear model $y = 5x + 10$ may describe a measured quantity that increases at an approximately constant rate. At $x = 6$, $y = 5(6) + 10 = 40$. In data analysis, a line can be fitted to observations to summarize a trend, although predictions outside the observed range may be unreliable.

B. Parabola: quadratic regression

A quadratic model can approximate data with a curved trend. For $y = x^2 - 4x + 7$, at $x = 3$, $y = 9 - 12 + 7 = 4$. The shape can represent a trend that decreases and then increases, or the reverse depending on the sign of the quadratic coefficient.

C. Power curve: algorithm complexity

An algorithm's operation count may grow approximately as $T(n) = 2n^2$ for some quadratic-time examples. If $n = 10$, then $T(10) = 2 \times 10^2 = 200$ units of work. This is an illustrative operation-count model, not a timing guarantee. Actual execution time also depends on hardware, implementation and input structure.



Practical value: mathematical functions help interpret data and compare how quickly computational work grows as input size increases.

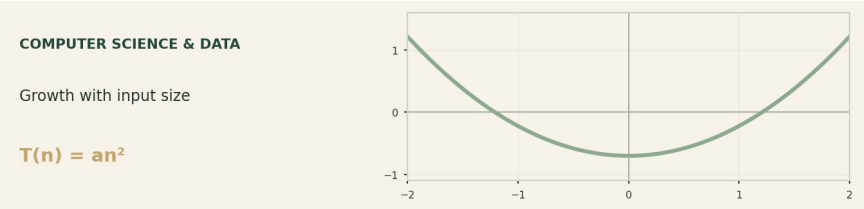


Figure: Illustrative model and graph — not experimental data.

COMPARISON AND FURTHER EXAMPLES

The three functions are useful for different patterns. Choosing the correct model begins by examining how the measured output changes as the input changes.

Straight-line model

Equation: $y = ax + b$. The slope a is constant. Examples include fixed-speed distance over time and a simplified cost model with a constant cost per item. A straight line is unsuitable if the rate of change varies substantially.

Parabolic model

Equation: $y = ax^2 + bx + c$. The graph has a vertex and a symmetric shape. It is useful for idealized projectile motion, quadratic optimization and curved trends over a limited range. If a is positive the vertex is a minimum; if a is negative it is a maximum.

Power-function model

Equation: $y = ax^b$. It can describe scale relationships where a simple linear or quadratic model is not appropriate. The exponent controls how the output responds to changes in the input. The domain must be considered, especially for fractional exponents.

Worked comparison

Let $x = 2$. For $y = 3x + 1$, $y = 7$. For $y = x^2 + 1$, $y = 5$. For $y = 2x^2$, $y = 8$. These different outputs show that the equations represent different relationships even when the same input value is used.

A good model should be simple enough to interpret, match the observed pattern, use consistent units and be tested against data not used to construct it.

INTERPRETATION, MODEL SELECTION AND LIMITATIONS

How to select a function

- Use a straight line when the rate of change is approximately constant over the range of interest.
- Use a parabola when a quadratic pattern is justified, such as ideal projectile motion or a curved relationship with a clear maximum or minimum.
- Use a power curve when data suggest a scaling relationship of the form $y = ax^b$.
- Plot the observations and inspect whether the model fits the data before using it for prediction.

Important limitations

Real-world systems are more complex than simple equations. A model may omit variables such as weather, human behaviour, measurement error, changing costs or equipment differences. The values of a, b and c must have appropriate units and be based on evidence when the model is intended for practical use.

A model that fits one range may fail outside that range. Extrapolation should therefore be done carefully. In medicine, transportation and engineering, safety-critical decisions require domain-specific standards and expert review.

Overall finding

Straight lines, parabolas and power curves are not competing formulas; they are different tools for describing different patterns. Their value lies in choosing a suitable equation, calculating correctly and interpreting the result within the context of the problem.

CONCLUSION AND FUTURE SCOPE

Conclusion

This project examined the applications of straight lines, parabolas and power curves in seven fields: medicine and healthcare, sports, economics and business, engineering and architecture, transportation, agriculture and environmental science, and computer science. Straight lines model constant rates of change, parabolas describe quadratic relationships and optimization patterns, and power functions represent scaling relationships.

The worked examples show how mathematical equations can convert information into useful estimates and help explain real-world patterns. They also demonstrate the importance of interpreting results carefully. The usefulness of a model depends on its assumptions, the quality of the data and the range in which it is applied.

Future Scope

- Collect real measurements and fit the equations using spreadsheets or statistical software.
- Compare linear, quadratic and power models to determine which best represents a particular dataset.
- Create interactive graphs to study how changing the constants changes the curve.
- Extend the project to exponential, logarithmic and trigonometric models used in other scientific applications.
- Investigate the accuracy of predictions by comparing model results with measured values.

REFERENCES / BIBLIOGRAPHY

The following are suitable starting points for studying functions, graphs, motion and mathematical modelling. Record the edition and access date for any source actually consulted.

1. National Council of Educational Research and Training (NCERT). Mathematics textbooks covering coordinate geometry, functions and quadratic equations.

2. OpenStax. Algebra and Trigonometry. Sections on functions, linear functions, quadratic functions and power functions.

3. OpenStax. College Physics. Sections on motion in two dimensions and projectile motion.

4. NIST/SEMATECH. e-Handbook of Statistical Methods. Material on regression and model fitting.

5. Stewart, James, Lothar Redlin and Saleem Watson. Precalculus: Mathematics for Calculus. Topics on functions and graphs.

6. Introductory materials in biological scaling and allometric relationships from academic textbooks or peer-reviewed research.

7. Introductory computer science textbooks covering algorithm analysis and time complexity.

Note: Add the exact edition, publisher details, URLs and dates accessed for the specific books or webpages used in the final submitted version.

APPENDIX

Appendix A: Key formulas

- Straight line: $y = ax + b$; slope = a; y-intercept = b.
- Quadratic function: $y = ax^2 + bx + c$, where $a \neq 0$.
- Vertex x-coordinate of a parabola: $x = -b/(2a)$.
- Power function: $y = ax^b$.
- Constant-speed motion: distance = speed × time.
- Idealized braking distance under constant deceleration: $d = v^2/(2a)$, where a is the magnitude of deceleration.

Appendix B: Glossary

Variable: a symbol representing a quantity that may change. Constant: a fixed value in a particular model. Slope: change in y divided by change in x for a straight line. Vertex: the turning point of a parabola. Model: a simplified mathematical representation of a real situation. Exponent: the power to which a quantity is raised.

Appendix C: Graphs

The graphs included in the applications section are illustrative examples. Their axes and equations should be labelled clearly if redrawn by hand.

Student notes / additional calculations:

_____

_____

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